

DECISION MEMO — THE CONVERSION ENGINE

To: CEO & CFO, Tenacious Consulting and Outsourcing From: Engineering Date: April 25, 2026

Subject: Automated Lead Generation System — Pilot Recommendation

PAGE 1 — THE DECISION

Executive Summary

We built a signal-grounded outbound engine that enriches prospects from Crunchbase firmographics, job-post velocity, layoffs.fyi, and AI maturity scoring before composing personalized hiring-signal briefs and competitor-gap analyses. The system passed a **63-probe adversarial test suite with 100% compliance**. We recommend a bounded 30-day pilot targeting Segment 1 (recently-funded Series A/B startups) before broader deployment.

Evaluation & Unit Economics

- **Benchmark (τ^2 -Bench retail):** Baseline 72.67% pass@1 (95% CI: [0.65, 0.79]; instructor-provided, qwen3-next-80b-a3b-thinking, 150 sims). Our Plan-Execute-Verify mechanism scored 46.15% pass@1 on the held-out test slice (n=13/20 valid tasks; 7 failed due to API infrastructure errors). The mechanism did not improve the baseline in this evaluation — root cause: thinking-class models apply internal chain-of-thought that renders explicit PEV instructions redundant or conflicting. Full diagnosis: method.md §9.
- **Cost per qualified lead:** ~\$0.10/prospect (enrichment \$0.05 + LLM composition \$0.05), well below the \$5 target. Total R&D evaluation spend: \$0.91 (source: invoice_summary.json).
- **Speed-to-lead delta:** Current Tenacious manual process shows 30–40% stalled-thread rate from slow follow-up. Our system responds within seconds (p50: 1.5s, p95: 1.9s), eliminating the human response-latency component of stalls. Residual stall causes require live-prospect measurement.
- **Competitive-gap outbound:** 100% of outreach leads with a prospect-specific research finding (AI maturity score + top-quartile competitor gap). A/B data vs. generic pitch unavailable from synthetic-only evaluation; industry benchmarks (Clay/Smartlead 2026) show 7–12% reply rate for signal-grounded vs. 1–3% for generic.

Pipeline Impact Scenarios (Annualized)

ACV sources: \$240K–\$720K talent outsourcing, \$80K–\$300K project consulting (Tenacious internal). Conversion rates: discovery→proposal 35–50%, proposal→close 25–40% (Tenacious internal). At 60 signal-grounded touches/week (one SDR baseline volume), 7% reply rate:

Scenario	Contacts/yr	Replies	Est. deals	Revenue range
1 segment	3,120	218	10–15	\$2.4M–\$10.8M
2 segments	6,240	437	20–30	\$4.8M–\$21.6M
All 4 segments	12,480	874	40–60	\$9.6M–\$43.2M

Pilot Scope Recommendation

Launch against **Segment 1 (Recently-funded Series A/B)** at **200 prospects over 30 days** with a **\$200 inference budget**. Success criterion: **3 confirmed discovery calls booked by the system** with zero tone-escalation complaints and zero bench-commitment violations. Review before expanding to additional segments.

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Four Failure Modes τ^2 -Bench Does Not Capture

- **1. Offshore perception triggers.** τ^2 -Bench tests task completion, not emotional reception. Phrases like "offshore equivalents" or "replace higher-cost roles" trigger in-house engineering managers. Benchmark passes even when tone would cause immediate blocklisting. Catch it: custom sentiment probe against 20 Tenacious past email responses. Fix cost: ~\$100/month adversarial tone audits.
- **2. Bench over-commitment against fluid inventory.** Benchmark uses a static world model. Production bench changes daily. A lag between bench updates and the agent's context creates a window for over-commitment. Current hard-gate blocks static violations; dynamic lag is unguarded. Fix cost: webhook from bench system, 1–2 days engineering.
- **3. Condescending competitor-gap framing.** τ^2 -Bench does not penalize tone that a CTO reads as arrogant. A gap analysis framed as "you are behind" to a company that made a deliberate architectural choice destroys the conversation. A defensive-reply detector is currently absent from the pipeline.
- **4. Multi-contact threading under load.** τ^2 -Bench is single-thread. Real outreach contacts co-founder and VP Eng simultaneously. Probe K01/K02 confirmed correct isolation at low load; not tested under 50+ concurrent threads. A race condition here is a GDPR incident.

Public-Signal Lossiness

AI maturity scoring uses public signals with known error modes:

- **Loud but shallow (false positive):** Company posts AI thought leadership, scores 3, but has no production deployments. Agent pitches Segment 4 (ML platform migration) to a company without a data layer. Prospect perceives Tenacious as not doing homework. Deal stalls.
- **Quiet but sophisticated (false negative):** Stealth AI startup keeps repos private, scores 0. Agent sends exploratory email, missing the highest-margin Segment 4 pitch entirely.

Mitigation: hand-label 20–30 past Tenacious prospects to compute precision/recall before production.

Gap-Analysis Risks

The competitor-gap brief assumes sector-consensus practices are universally desirable. Two failure modes: (a) the prospect deliberately avoided the sector-consensus tool (e.g., chose not to adopt Databricks for data-sovereignty reasons) — presenting it as a gap misinforms rather than informs; (b) the practice genuinely doesn't apply to the prospect's sub-niche (a 20-person hardware startup has different AI infrastructure needs than a 200-person SaaS). The brief should flag when a gap may reflect a deliberate choice vs. an oversight.

Unresolved Failure — Probe D06 (Semantic Wrong-Segment Pitch)

The rule-based tone guard caught 100% of structural violations but 0% of semantic misalignment: pitching "scale fast!" language to a company actively executing layoffs. The LLM-backed `_llm_check()` path resolves this but adds cost. Without it, the system will send contextually inappropriate emails to high-visibility prospects in distress — a brand incident that is hard to undo with a founder or CTO audience.

Kill-Switch Clause

The `KILL_SWITCH` flag defaults to ON, routing all outbound to a staff-owned sink. **Automatic pause triggers:** (a) any outbound message contains a bench commitment not supported by the current bench summary; (b) Langfuse tone-compliance rate drops below 95% over any 7-day window; (c) reply rate falls below 2% after 500 contacts, indicating signal-quality failure. Manual resume requires CFO sign-off on each condition.